

Demonstrate the Current AI Workflow

Project: HisabDo Website Enhancement

Department: AI/ML

Module: HisabDo AI Copilot

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1. Introduction

The purpose of this document is to demonstrate the current end-to-end workflow of the proposed **HisabDo AI Copilot**.

The AI system is designed to process different types of user requests in a controlled and structured manner. Instead of sending every user question directly to an LLM, the system first analyzes the request, detects its intent, identifies the required information, and routes it to the appropriate AI or application module.

The current workflow is based on the following principle:

AI understands and explains. Verified systems retrieve, calculate, and decide.

This approach is particularly important because HisabDo handles financial information such as customer balances, transactions, expenses, receivables, and other sensitive business data.

The system therefore separates:

- General AI conversations
- Product and feature questions
- FAQ requests
- Knowledge retrieval
- Financial queries
- Business insights
- Recommendations

Each request follows an appropriate processing path.

2. Current AI Workflow Overview

The complete workflow of the HisabDo AI Copilot is:

User Query

↓

Input Processing



Language Detection / Input Normalization



Intent Detection



AI Orchestrator



Request Classification and Routing



Knowledge / FAQ / Financial / Insight Module



Verified Context or Verified Data



Controlled Response Generation



Response Validation



User Response

The AI Orchestrator acts as the central decision-making layer responsible for determining which module should process the request.

3. End-to-End AI Request Flow

When a user submits a message, the system should not immediately generate a response.

Instead, the request passes through multiple controlled stages.

USER



USER SUBMITS QUERY

RESPONSE VALIDATION

|



USER RESPONSE

4. Step 1 — User Query

The workflow begins when a user submits a question through the HisabDo AI interface.

The query may be entered through:

- AI Chatbot
- Website Chat Interface
- Dashboard Assistant
- Voice Interface
- AI Suggestion Panel
- Help and Support Section

Example user queries:

What is HisabDo?

How do I add a customer?

Ali ka kitna udhaar hai?

How can I restore my backup?

Mere is month ke expenses kitne hain?

What should I review today?

Each query may require a completely different processing path.

5. Step 2 — Input Processing

Before the AI system analyzes the request, the input should be processed and validated.

The input processing layer performs basic preparation.

Input Processing Responsibilities

- Validate the request
- Check whether the message is empty
- Remove unnecessary formatting
- Normalize excessive spaces
- Handle unsupported characters
- Detect obvious input errors
- Prepare text for further AI processing

Example:

Original Input:

Ali ka kitnaaaa udhar hai???

After normalization:

Ali ka kitna udhar hai?

The system should preserve the original meaning while making the input easier for downstream AI modules to process.

6. Step 3 — Language Detection and Input Normalization

HisabDo AI Copilot is designed to support multilingual interaction.

The system should identify the user's language before processing the request.

Initial supported languages include:

- English
- Urdu
- Roman Urdu
- Hindi
- Arabic

Example:

User Input:

Ali ka kitna udhaar hai?

Detected language:

Roman Urdu

Another example:

User Input:

میرے اس مہینے کے اخراجات کتنے ہیں؟

Detected language:

Urdu

The system should generally follow:

Input Language



Language Detection



Process Request



Generate Response



Respond in Same Language

7. Roman Urdu Input Normalization

Roman Urdu requires additional processing because users may write the same word in different forms.

Examples:

udhar

udhaar

udharr

kitna

kitny

kitnay

paisa

paisay

pese

The normalization layer should identify these variations and map them toward a consistent understanding.

Example:

Ali ka kitny pese rehndy hain?

Possible normalized meaning:

Ali ka kitna paisa outstanding hai?

The goal is not necessarily to force users to write in a standard format.

Instead, the system should understand natural user input.

8. Step 4 — Intent Detection

After processing the input, the system identifies what the user wants.

This process is called **Intent Detection**.

Example:

User Query:

Ali ka kitna udhaar hai?

The system may classify it as:

Intent:

CUSTOMER_BALANCE_QUERY

Entity:

Ali

Request Type:

Financial Query

Required Data:

Customer Balance

Response Type:

Verified Financial Response

Another example:

User Query:

How do I export a PDF?

Classification:

Intent:

FEATURE_GUIDANCE

Request Type:

Product Help

Required System:

Knowledge Base / FAQ / RAG

Response Type:

Documentation-Based Response

9. Initial Intent Categories

The current AI workflow can support the following intent categories:

PRODUCT_INFORMATION

FAQ_QUERY

FEATURE_GUIDANCE

CUSTOMER_BALANCE_QUERY

TRANSACTION_QUERY

EXPENSE_QUERY

RECEIVABLE_QUERY

PAYABLE_QUERY

BUSINESS_INSIGHT

ANALYTICS_QUERY

RECOMMENDATION_QUERY

REPORT_HELP

BACKUP_HELP

LANGUAGE_SUPPORT

GENERAL_CONVERSATION

UNKNOWN

These intent categories allow the AI Orchestrator to route each request to the correct processing module.

10. Step 5 — AI Orchestrator

The **AI Orchestrator** is the central routing component of the proposed HisabDo AI architecture.

Its responsibility is to determine:

What type of request is this?

What information is required?

Which system should handle the request?

What type of response should be generated?

The AI Orchestrator does not necessarily answer every question directly.

Instead, it coordinates the appropriate modules.

User Request



Intent Detection



AI Orchestrator



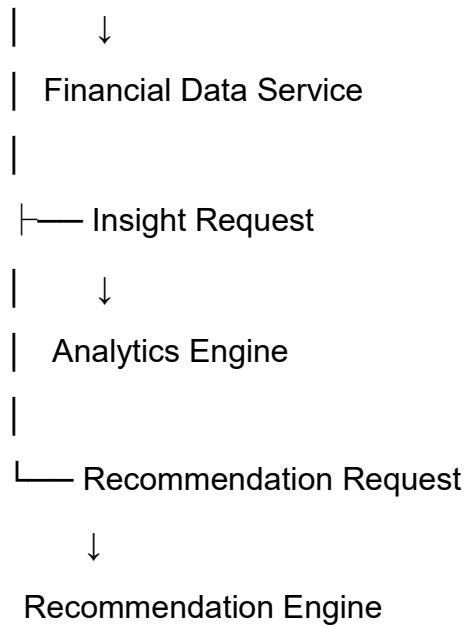
| — Product Question



| RAG / FAQ



| — Financial Question



11. Product and Help Query Handling

Product-related questions should use the **Knowledge Intelligence Layer**.

Example:

User:

How do I add a customer?

Processing flow:

User Query



Input Processing



Language Detection



Intent Detection



FEATURE_GUIDANCE



AI Orchestrator

↓
Knowledge Base Search

↓
Relevant Documentation

↓
Verified Context

↓
Response Generation

↓
User Response

The system should retrieve relevant HisabDo information before generating the final answer.

12. Knowledge Base and FAQ Search Flow

The Knowledge Base contains verified product information.

Possible sources include:

FAQs

+

Product Documentation

+

Feature Guides

+

Help Articles

+

Website Content

+

Blog Content

+

Product Descriptions

The AI workflow should follow:

User Question



Knowledge Search



Relevant FAQ / Document Search



Retrieve Matching Content



Verified Context



AI Response Generation



Final Answer

The AI should be grounded in available HisabDo information.

13. RAG Workflow

For advanced knowledge retrieval, the system uses **Retrieval-Augmented Generation (RAG)**.

The current conceptual workflow is:

HisabDo Knowledge Base



Document Cleaning



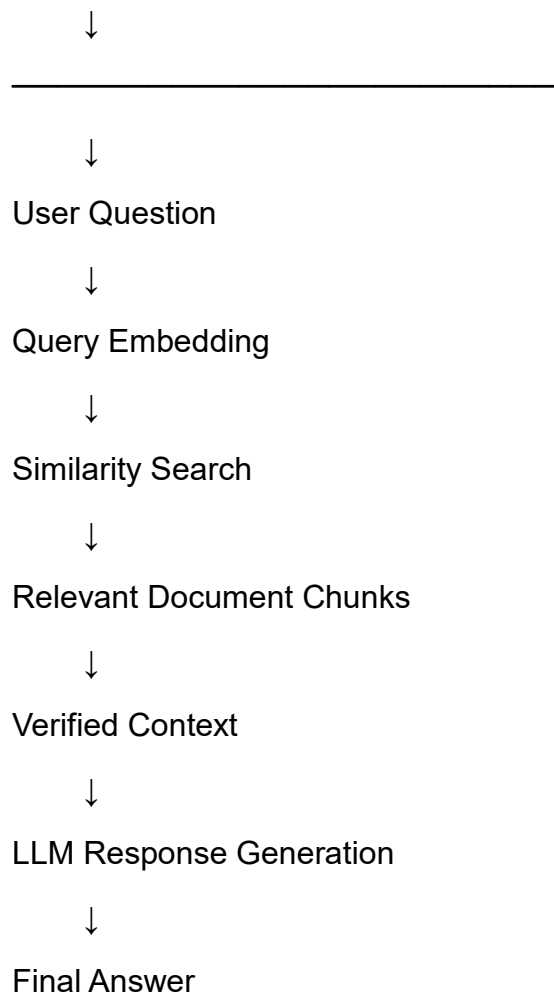
Text Chunking



Embedding Generation



Vector Database



The core principle is:

Retrieve Relevant Knowledge

↓

Provide Verified Context

↓

Generate Controlled Response

14. Step 6 — Financial Query Handling

Financial queries require a separate and more secure workflow.

Example:

User:

Ali ka kitna udhaar hai?

The system should not allow the LLM to invent or independently calculate the answer.

The workflow should be:

User Query



Intent Detection



CUSTOMER_BALANCE_QUERY



Entity Extraction



Customer = Ali



AI Orchestrator



Financial Data Service



Verified Application API



Application Database



Business Logic



Verified Financial Result



Response Generation



User Response

15. Financial Data Example

The Financial Data Service may return structured information.

Example:

```
{  
  "customer": "Ali",  
  "balance": 25000,  
  "currency": "PKR",  
  "source": "verified_application_data"  
}
```

The AI response layer receives verified structured data.

The AI can then generate a natural response:

Ali ka current outstanding balance PKR 25,000 hai.

The important distinction is:

Financial Value:

Verified System

Language Explanation:

AI

16. Verified Financial Data Architecture

Financial requests should follow this architecture:

USER

↓

AI UNDERSTANDS THE QUESTION

↓

AI IDENTIFIES REQUIRED DATA

↓

CONTROLLED API REQUEST

↓

APPLICATION BUSINESS LOGIC



DATABASE



VERIFIED RESULT



AI EXPLAINS RESULT



USER

The LLM should not:

- × Directly access MongoDB
- × Write arbitrary database queries
- × Invent balances
- × Estimate financial values
- × Access another user's data

Instead:

LLM



Controlled Tool / API



Authorized Application Service



Verified Data

17. General Query vs Financial Query

General Product Question

Example:

How does backup work?

Workflow:

User



Intent Detection



BACKUP_HELP



AI Orchestrator



RAG / Knowledge Base



Relevant Documentation



AI Response

Financial Query

Example:

Ali ka balance kitna hai?

Workflow:

User



Intent Detection



CUSTOMER_BALANCE_QUERY



AI Orchestrator



Financial Data Service



Verified API



Business Logic



Verified Result



AI Explanation

This separation is essential for the accuracy and safety of HisabDo AI Copilot.

18. Step 7 — Verified Context

Before the final response is generated, the AI system should provide verified context to the response layer.

The source of context depends on the request type.

Product Question



Knowledge Base Context

Financial Question



Verified Application Data

Insight Question



Analytics Results

Recommendation Question



Rules / User Context / Analytics

The AI response should be based on this controlled context whenever factual information is required.

19. Step 8 — Controlled Response Generation

The Response Engine receives:

User Question

+

Detected Language

+

Detected Intent

+

Verified Context

+

Response Instructions

The AI then generates a user-friendly answer.

Example:

Structured Data:

Customer: Ali

Balance: 25000

Currency: PKR

Generated response:

Ali ka current outstanding balance PKR 25,000 hai.

The AI is responsible for making the answer natural, understandable, and language-appropriate.

20. Response Validation

Before returning the response, the system should perform basic validation.

The validation layer should check:

- Whether a response was generated
- Whether verified context was available when required
- Whether financial data is consistent

- Whether the response belongs to the correct language
- Whether an unsupported claim was generated
- Whether sensitive information is exposed
- Whether the response should use a fallback message

Workflow:

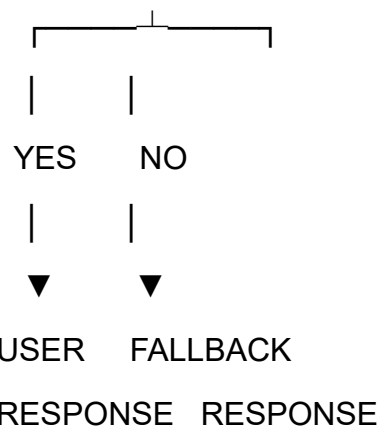
Generated Response



Validation Layer



Valid?



21. Fallback Behavior

The AI system must handle situations where reliable information is unavailable.

Knowledge Fallback

User Query



Knowledge Search



No Reliable Information Found



Safe Fallback Response

Example:

I could not find enough verified information to provide an accurate answer. Please try rephrasing your question.

Financial Fallback

If verified financial data cannot be retrieved:

Financial Request



Financial Data Service



Data Unavailable



DO NOT ESTIMATE



Safe Error Response

Example:

I could not retrieve the current verified financial information at the moment. Please try again later.

The system must never generate a guessed financial value.

22. Sample Demonstration Scenario 1 — Product Information

User Query

What is HisabDo?

Processing

Input Processing



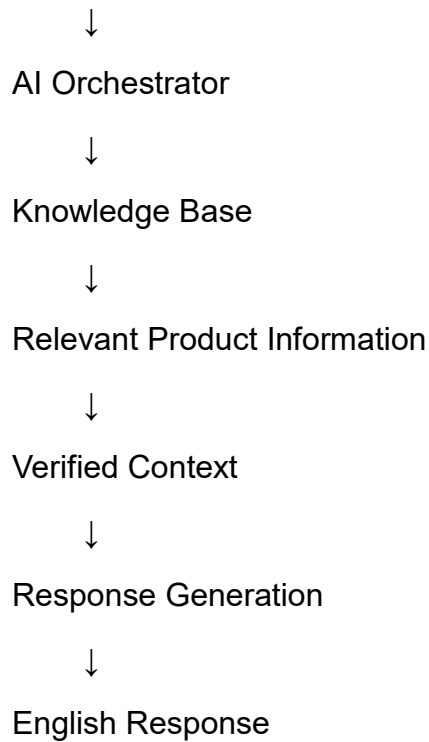
Language Detection: English



Intent Detection



PRODUCT_INFORMATION



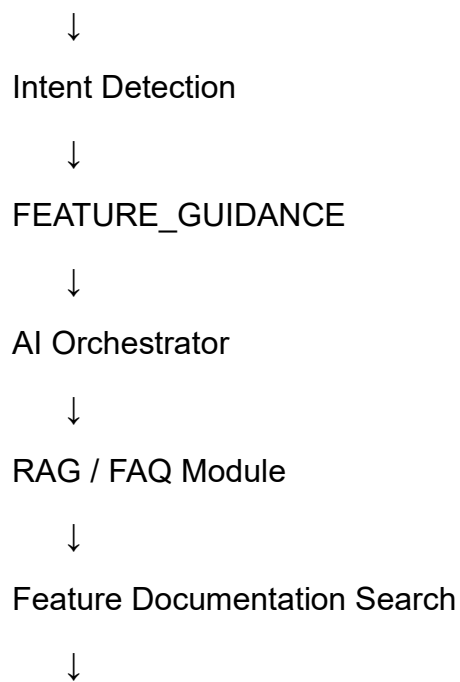
23. Sample Demonstration Scenario 2 — Feature Guidance

User Query

How do I add a customer?

Processing

User Query



Relevant Instructions



Response Generation



User Response

24. Sample Demonstration Scenario 3 — Financial Query

User Query

Ali ka kitna udhaar hai?

Processing

User Query



Language Detection



Roman Urdu



Intent Detection



CUSTOMER_BALANCE_QUERY



Entity Extraction



Ali



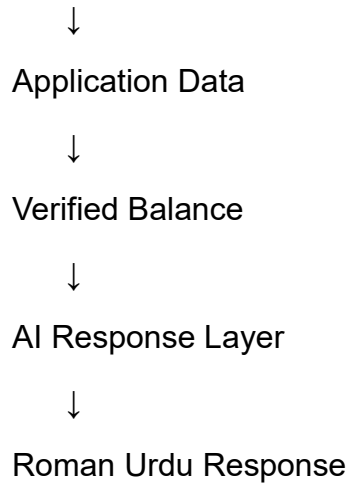
AI Orchestrator



Financial Data Service



Verified Customer API



Example final response:

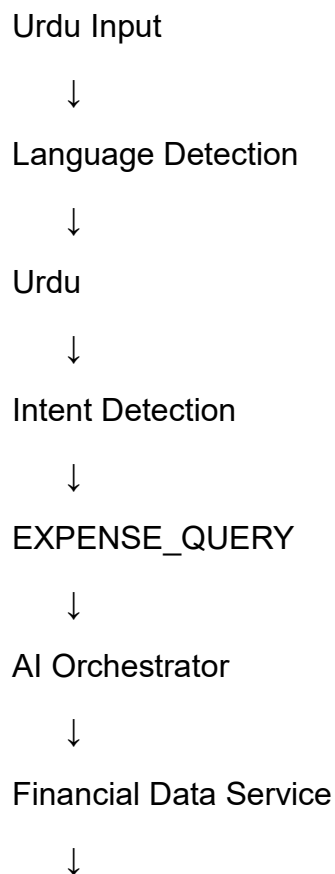
Ali ka current outstanding balance PKR 25,000 hai.

25. Sample Demonstration Scenario 4 — Urdu Financial Query

User Query

میرے اس مہینے کے اخراجات کتنے ہیں؟

Processing



Verified Expense Data



Business Logic



Structured Result



AI Response



Urdu Output

26. Sample Demonstration Scenario 5 — AI Recommendation

User Query

What should I review today?

Processing

User Query



Intent Detection



RECOMMENDATION_QUERY



AI Orchestrator



Recommendation Engine



User Context

+

Analytics Data

+

Business Rules

↓
Relevant Recommendation

↓
AI Explanation

↓
User Response

Example:

You have three customers with pending balances. Reviewing their payment status may be a useful next step.

27. Current AI Workflow Summary

The proposed HisabDo AI workflow can be summarized as:

USER

↓
INPUT PROCESSING

↓
LANGUAGE DETECTION

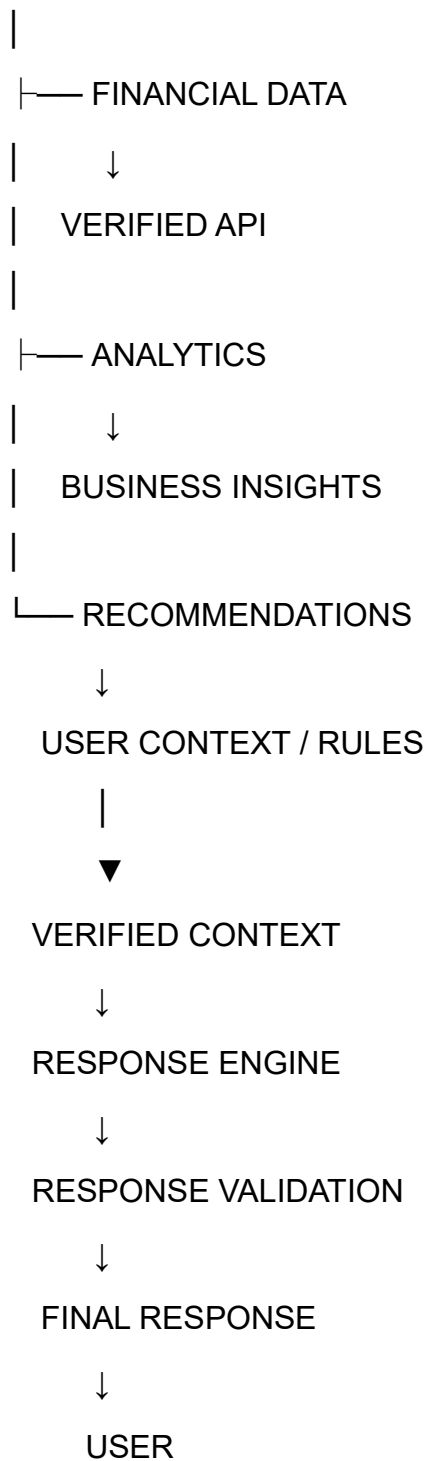
↓
INPUT NORMALIZATION

↓
INTENT DETECTION

↓
AI ORCHESTRATOR

↓
REQUEST ROUTING

|
| — KNOWLEDGE / FAQ
| ↓
| RAG

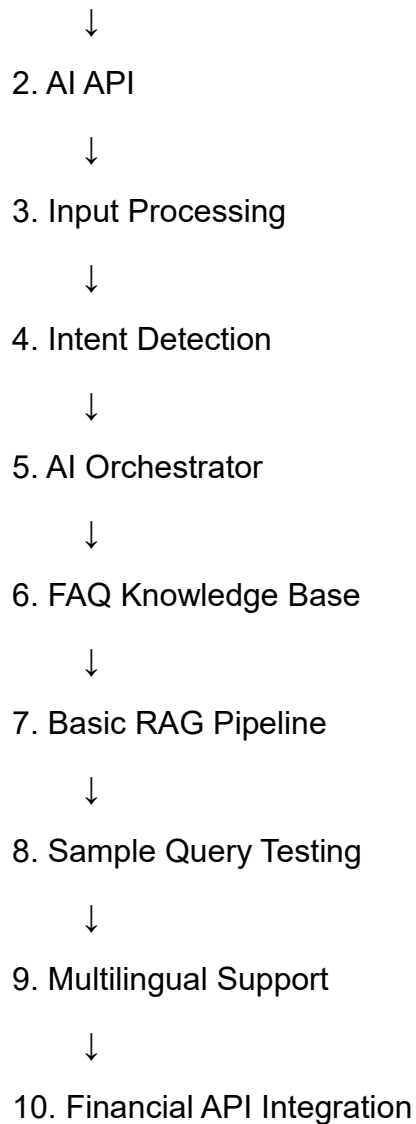


28. Current Workflow Status

The current workflow should be treated as the **core architecture and implementation blueprint** for the HisabDo AI Copilot.

The immediate implementation priority should be:

1. Basic Chat Interface



Advanced features such as recommendations, insights, voice AI, predictive intelligence, and offline AI optimization can be introduced in later phases.

29. Final Architecture Principle

The complete AI workflow follows one core rule:

AI Understands

+

AI Routes

+

Verified Systems Retrieve

+

Business Logic Calculates

+

AI Explains

For general questions:

Knowledge



RAG



AI Response

For financial questions:

Verified Application Data



Business Logic



Verified Result



AI Explanation

This architecture ensures that HisabDo AI Copilot is not dependent on an LLM as the source of truth.

Instead, it combines:

- AI understanding
- Controlled orchestration
- RAG-based knowledge retrieval
- Verified application data
- Secure API access
- Deterministic business logic
- Multilingual interaction
- Response validation
- Safe fallback handling

Conclusion

The proposed current AI workflow demonstrates how the **HisabDo AI Copilot** can process different categories of user requests through a controlled, modular, and scalable architecture.

The system does not treat every user query in the same way. Instead, it first understands the request and then routes it to the correct intelligence layer.

Product Question



Knowledge / RAG

Financial Question



Verified Financial Service

Business Insight



Analytics Engine

Recommendation



Recommendation Engine

The most important architectural principle remains:

AI should understand the user's question and explain verified results. The AI should not become the source of truth for financial information.

This workflow provides a practical foundation for implementing the HisabDo AI Copilot in future development phases while maintaining **accuracy, privacy, security, multilingual support, scalability, and user trust**.

HisabDo AI Copilot – Complete AI Workflow

End-to-End Workflow from User Query to Verified Response

